

Using Edge Computing with Drones to Optimize Crop Yield Prediction

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Abstract

Improving crop yield prediction is a significant problem in today's society. When farmers can predict the yield of their crops with improved efficiency, they can take the necessary steps to improve the yield of their crops and crop fields. By doing so, they can improve agricultural efficiency, food security, and sustainability. Although there are methods to predict crop health, most of which use machine learning (ML) models, they have significant drawbacks. Primarily, the high latency and delay between uploading an image or sensor data and receiving the desired crop health information. Farmers need consistent, real-time access to this information to effectively monitor their crops. This paper proposes a solution that attempts to address these drawbacks through an edge computing framework in which an ML model is run directly on a drone. In this way, a drone can take a picture of a corn crop field, process the image, run the model, and output the desired information in an efficient process that minimizes delays. This paper explores using multiple ML algorithms to address this task, and the model will take in hyperspectral imagery as input. This paper aims to present the solution of a fully functioning ML model integrated into a drone. For this, two ML models, a Deep Neural Network (DNN) and a Gradient-Boosted Regression Tree (GBRT), were built, alongside a script to read in and process hyperspectral imagery. When combined, this created a crop yield prediction system with comparable performance to existing models, while running within the computing limitations of common edge computing devices that can be integrated into real-world drones.

1 Introduction

Precision agriculture is the use of various technologies such as GPS, sensors, and automation to improve the efficiency of farms (GAO, 2024). One of the major precision agriculture techniques is the analysis of agricultural imagery to predict crop yield. The most common of these techniques for crop yield prediction today is through machine learning (ML), specifically computer vision (CV). Through this method, a machine learning model is trained on aerial and/or satellite images of crop fields and information about the respective crop yield. When an image of a new crop field is then uploaded to a given model, most likely

taken by a drone, a prediction can be made about its forecasted crop yield, and farmers can take steps to maximize the yield of that given crop field.

One specific type of imagery that is especially useful for crop yield prediction is hyperspectral imagery. Taken with specialized cameras, these images capture the entire continuous spectrum of visible light, and even a portion of the infrared spectrum. In contrast, regular cameras only capture red, green, and blue light (RGB).

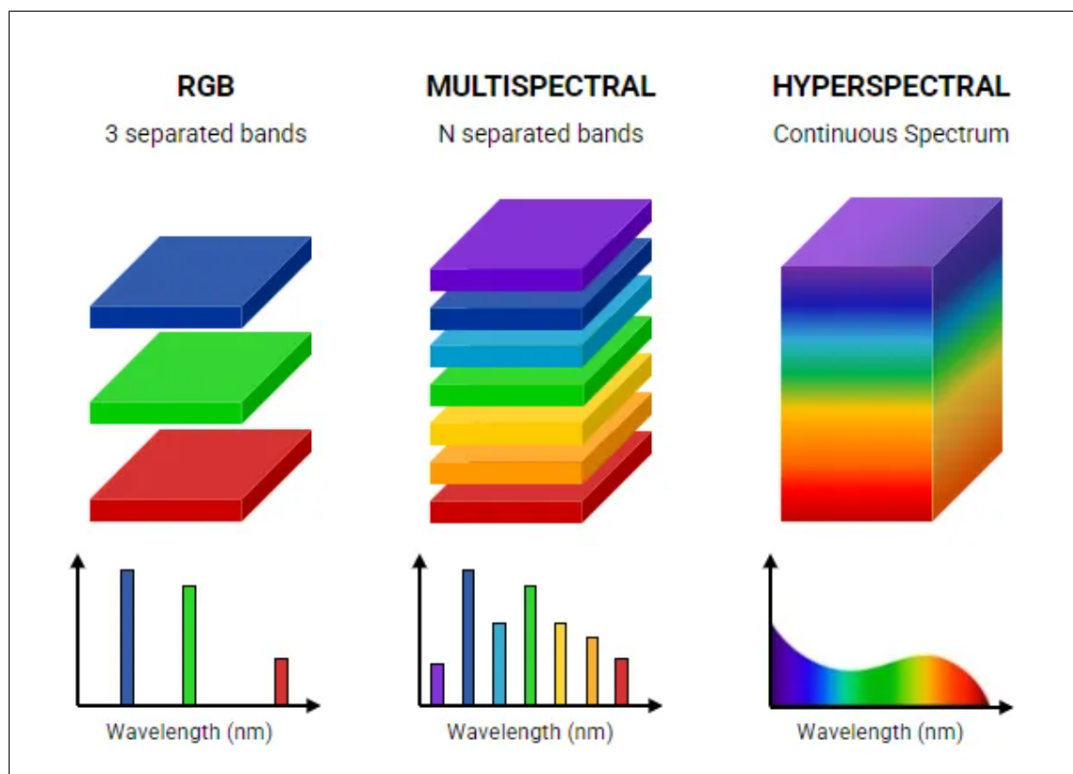


Figure 1: RGB vs Multispectral vs Hyperspectral Imaging (NIREOS, 2024)

The distinct portions of the spectrum that hyperspectral images capture are known as bands, which are classified based on wavelength ranges and are generally separated into the Red, Blue, Green, Red-Edge, and Near-Infrared bands. Hyperspectral imaging can reveal a lot more information about the state of crops, including moisture levels, soil nutrient levels, and presence of diseases/pests (GeoPard Agriculture). With crop yield prediction specifically, hyperspectral images can extract information about crop field health through vegetation indices (VIs). These are metrics that take into account factors such as color, moisture, and chlorophyll levels to assess the health and condition of crops in a given field. Because crop health correlates directly with crop yield, crop health information through VIs can be used with a predictive model to determine yield.

The prediction of crop health today through ML has two major drawbacks, however: the latency and delays between processing data and acquiring crop health information, and the need for a high-end device with adequate computing power to run a predictive model.

Primarily, without consistent, reliable internet access, farmers have trouble conducting crucial precision agriculture tasks, which have the potential to boost crop yields (Vittekk, 2022). Further, predictive models, specifically computer vision models, have specific computational power requirements, often requiring a dedicated high-end device to run models (Thompson et. al, 2020).

One important way in which issues regarding high latency in machine learning (ML) are addressed is through edge computing. This is the process of bringing applications closer to their initial source of data, specifically edge servers or IoT devices, with the goal of faster response times and reduced bandwidth (IBM). Through this project, an edge computing solution will be created, in which an ML model is run directly on a drone using an attached edge computing device. With this, a drone can take a picture of a crop field, process the image, run the ML model, and output the desired information in an efficient process that minimizes delays and doesn't require a high-end phone or laptop to process images and run the model.

To train and test the ML models in this solution, tabular data will be used in the form of VI information extracted from hyperspectral imagery of crop fields, along with the respective yield. Further, DNN and GBRT algorithms will be the framework behind the models, as they are commonly used for tabular data and provide high-accuracy predictions.

To show that the system can run directly on an edge computing device such as the DJI Manifold or NVIDIA Jetson, the computational requirements of the system will be tested. Specifically, the RAM usage will be tested and shown to run using under 8 GB of RAM, which is the upper memory limit for most common edge computing devices.

This project has real-world significance because it offers a way for farmers around the world to improve the efficiency with which they analyze and treat crops. This allows them to take the necessary actions to maximize crop yield, which can include changing water levels, nutrient levels, and types of fertilizers and pesticides used. By improving crop yield, farmers can increase agricultural efficiency and economic output globally.

There is existing research on the different techniques used in the methodology, but no research using the exact solution that will be created. For example, there is a solid amount of research on using computer vision models with aerial and satellite imagery for precision agriculture. Papers such as Computer vision in smart agriculture and precision farming: Techniques and applications and Deep Learning Models for the Classification of Crops in Aerial Imagery: A Review are examples of the various utilizations of ML and computer vision for precision agriculture tasks such as crop health prediction, crop classification, and other smart agriculture tasks. While the solutions discussed in these papers still fall victim to the drawbacks discussed, this research is still extremely useful to this project. The techniques used in these existing projects can be leveraged when creating models for this project. Furthermore, there is research on running ML models on drones, which can be referenced to run the desired models on drones (Sriram, 2023).

However, while existing research provides solutions to crop yield prediction, none of them look at the issue of latency and delays for making predictions, especially for farmers in remote areas with low internet access. Further, there is no proposition of an edge computing solution

to address latency issues for this specific field, which is what this paper proposes.

2 Methodology

2.1 Predictive Models

The first step to developing this solution was to create an ML model that could accurately predict crop yield given VI data. This process started by first collecting data that contained hyperspectral imagery, the corresponding crop field health markers in the form of VIs, and respective crop yield. This data was collected from "Forecasting spring maize yield using vegetation indices and crop phenology metrics from UAV observations", a paper in which a similar crop yield prediction project is completed (Bao et al., 2023). This data included VI information from 60 sample plots of land across two corn fields, as well as data from seven different stages of growth. Because real-world predictions would be made on crops from a singular stage of growth, the dataset was split based on growth stage such that predictive models could be trained with data from each stage separately. An image of the first crop field during the fourth growth stage, as well as a section of the dataset, are shown below (Bao et al., 2023).



Figure 2: Corn Crop Field 1 During Growth Stage 4

ID	NDVI1	EVI1	SR1	MTCI1	GNDVI1	NDRE1
1	0.85655	0.549745	13.58736	1.224404	0.752551	0.338998
2	0.80874	0.482898	10.83421	1.183232	0.722481	0.316661
3	0.829015	0.501701	11.60439	1.181634	0.734735	0.322916
5	0.850841	0.559445	13.45338	1.220911	0.748847	0.335924
8	0.814347	0.435275	10.91362	1.098211	0.710683	0.304203
10	0.889131	0.644502	17.75425	1.435598	0.791797	0.382458
11	0.874352	0.635166	15.6881	1.28038	0.771198	0.354434
13	0.875349	0.58502	15.60303	1.291674	0.776125	0.356153
14	0.876439	0.670829	15.90177	1.296207	0.772724	0.35734
16	0.806771	0.393835	10.30854	1.211844	0.714708	0.32027

Table 1: Vegetation Indices Data

Following data collection, two machine learning models were built, using a DNN algorithm and GBRT algorithm, respectively. These algorithms in specific were chosen due to their ability to efficiently handle tabular data, and a GBRT algorithm is common for crop yield prediction tasks in existing research such as the paper mentioned above. These models were built, trained, and tested using Python, through a TensorFlow framework with Keras. The were trained on the aforementioned hyperspectral data, trained and tested with data from each growth stage. The performance of these models were evaluated through their mean absolute error, and hyperparameters such as batch size an epochs were tuned to maximize performance. The model performance of both models through each of the seven growth stages were compared.

2.2 Image Processing

After the completion of the predictive models, a script was created to process hyperspectral images to extract VI information that acts as the input data into the models. For this, Python was used with the Rasterio library to read in a given image. This outputted intensity values from each pixel for each hyperspectral band, through which VIs could be calculated using the following equations:

$$NDVI = \frac{NIR_band - Red_band}{NIR_band + Red_band} \quad (1)$$

$$EVI = \frac{2.5 \times (NIR_band - Red_band)}{NIR_band + 6 \times Red_band - 7.5 \times Blue_band + 1} \quad (2)$$

$$SR = \frac{NIR_band}{Red_band} \quad (3)$$

$$\text{GNDVI} = \frac{\text{NIR_band} - \text{Green_band}}{\text{NIR_band} + \text{Green_band}} \quad (4)$$

$$\text{CIGreen} = \frac{\text{NIR_band}}{\text{Green_band}} - 1 \quad (5)$$

$$\text{EVI2} = \frac{\text{NIR_band} - \text{Red_band}}{1 + \text{NIR_band} + 2.4 \times \text{Red_band}} \quad (6)$$

$$\text{MNDVI} = \frac{\text{NIR_band} - \text{Red_band}}{\text{NIR_band} + \text{Red_band} - 2 \times \text{Blue_band}} \quad (7)$$

$$\text{MSR} = \frac{\left(\frac{\text{NIR_band}}{\text{Red_band}} - 1\right)}{\sqrt{\left(\frac{\text{NIR_band}}{\text{Red_band}} + 1\right)}} \quad (8)$$

$$\text{DVI} = \text{NIR_band} - \text{Red_band} \quad (9)$$

Here, "Red_band", "Green_band", "Blue_band", and "NIR_band" represent the pixel intensities of the red, green, blue, and near-infrared bands, respectively. The red edge band could not be obtained from the data, so data that required the red edge band to be extracted was omitted from model training. Despite this, the models maintained performance.

2.3 Edge Computing

The final step was tuning the system to run using less than 8 GB of RAM, showing that it could run on most common edge computing devices. To do this, the Psutil Python library was used to constantly monitor RAM usage as the program ran. It was initially exceeding 8 GB of RAM due to the image processing. This step had to be altered by processing images in chunks rather than all at once, which allowed the system to run within 8 GB of RAM.

3 Results and Discussion

To assess the performance of both the Deep Neural Network (DNN) model and the Gradient-Boosted Regression Tree (GBRT) model, mean absolute error (MAE) was used. The equation to determine MAE is listed below:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - x_i| \quad (10)$$

Here, the absolute error between the actual and predicted output values, represented by y_i and x_i , respectively, is summed up and divided by the total number of predictions. The lower the MAE, the better the performance, as it indicates that the predicted values are closer to the true values. Below is a table of the validation MAE for both the DNN and GBRT models, for each stage of growth for the corn crops.

Growth Stage	DNN Validation Error (100kg/ha)	GBRT Validation Error (100kg/ha)
1	16.42	20.45
2	19.75	15.56
3	16.99	17.69
4	16.27	11.05
5	13.69	14.39
6	13.10	14.11
7	9.60	14.97

Table 2: Model MAE by Growth Stage

As seen in the table, the MAE for both models decreases for later growth stages, indicating better performance. This is likely due to the final yield being easier to predict as the crops grow more. Furthermore, the MAE in the later stages range from 960 - 1439 kg/ha, which is comparable to existing model performance of 900 - 1200 kg/ha, as shown in the paper that is associated with the dataset used.

To better understand and visualize the performance of both models, an example plot was created plotting the actual yield data against the predicted values for both models, for growth stage 1. This plot is shown below.

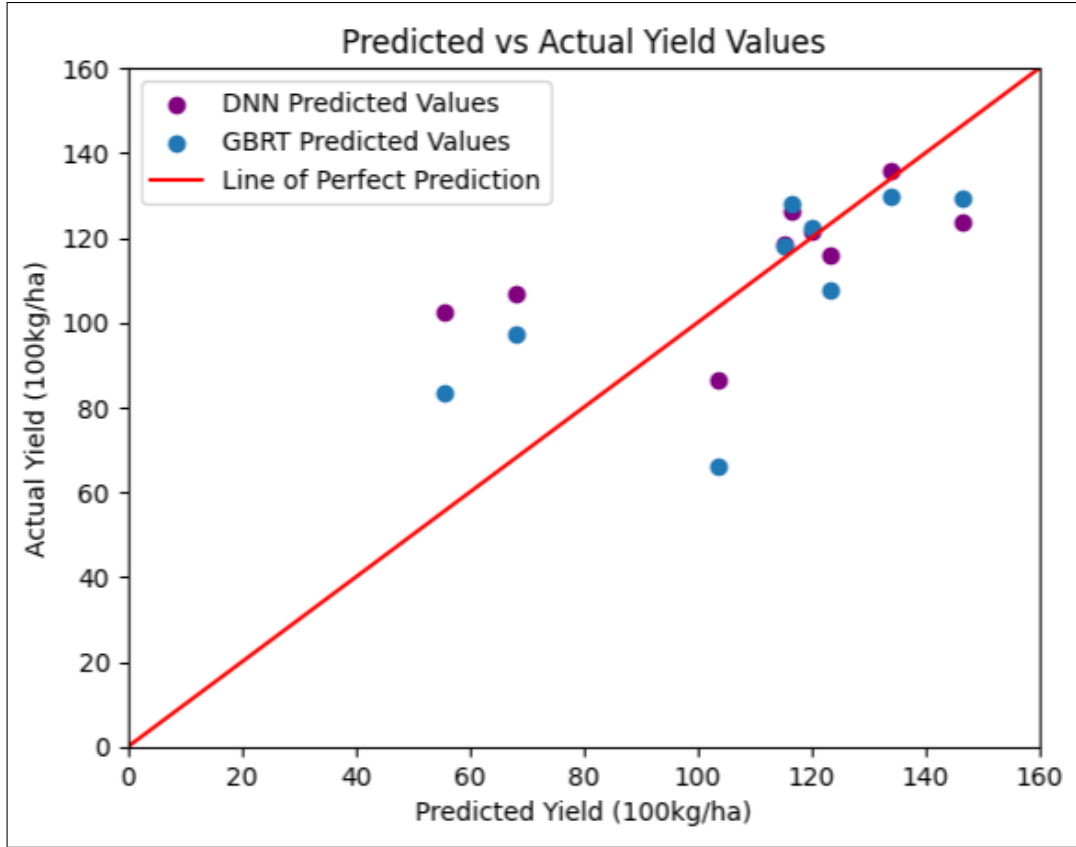


Figure 3: Predicted vs Actual Yield Values

In this plot, the blue and purple points represent the predictions made by the DNN and GBRT models, respectively. The red line is a perfect 1 to 1 line between predicted and actual values, meaning that the closer each data point is to the line, the better prediction it is. As shown by the plot, both models predicted yield for each sample relatively similarly. It can also be noted that both models make better predictions for higher true yield values. This can be explained by the higher frequency of samples with a higher yield, compared to those with lower yield. Because there is a better representation of higher yield plots, the models make better predictions when the validation data has a higher true yield.

Below is a log of memory usage for the image processing step, which was the most memory-intensive step.

```

Current memory usage: 5.60 GB
Current memory usage: 5.60 GB
Current memory usage: 5.76 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Current memory usage: 6.14 GB
Finished memory-intensive function.
Mean Vegetation Indices: [2.8388956e-01 8.6238239e+02 2.8051875e+00 2.5618106e-01 1.6136844e+00
1.3648471e-01 3.2059677e+01 7.3624057e-01 3.6133513e-01]
Program finished.

```

Figure 4: System Memory Usage for Image Processing

This portion was shown to complete using less than 8 GB of RAM. This shows that this system can run on a drone via an edge computing device such as DJI Manifold or NVIDIA Jetson, which are common devices used for ML tasks.

4 Conclusions

The design goals of this project were mostly met up to this point. ML models were created that can predict the final yield of a corn crop field with high accuracy, comparable to existing research. A script was also developed that can process hyperspectral images to extract VI data, which can be used in the ML models to make predictions. This system was shown to run within the computing limits of most common edge computing devices, specifically running within 8 GB of RAM.

A major drawback that limited model performance was the small sample size of the dataset. There were only two crop fields captured, resulting in only 60 rows of VI data, as 30 sample plots were taken from each field. However, taking this into account, the models still had solid performance.

Due to time and budget constraints, real field testing through the implementation of the system onto a drone and testing with real-world crop field images could not be performed. However, as the system was shown to be able to run on a drone with high performance, the majority of the design goals were still met.

The real-world applications of these conclusions are vast. Most importantly, this has implications worldwide for improving crop yield prediction by more quickly and efficiently delivering crop yield information. This will allow farmers, especially those in rural areas far from data centers, to learn how their crops will grow and what their yield will be in an efficient manner, allowing them to take the necessary action to increase crop yield.

4.1 Future Work

Based on time and budget constraints, future steps would be implementing the completed system onto a drone through an edge computing device, likely using a DJI drone with DJI Manifold. Field testing would then be performed to evaluate model accuracy based on real-world images taken by the drone. Possible further extensions include building a mobile app or user interface in which farmers could view and organize crop yield information outputted from the drone.

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